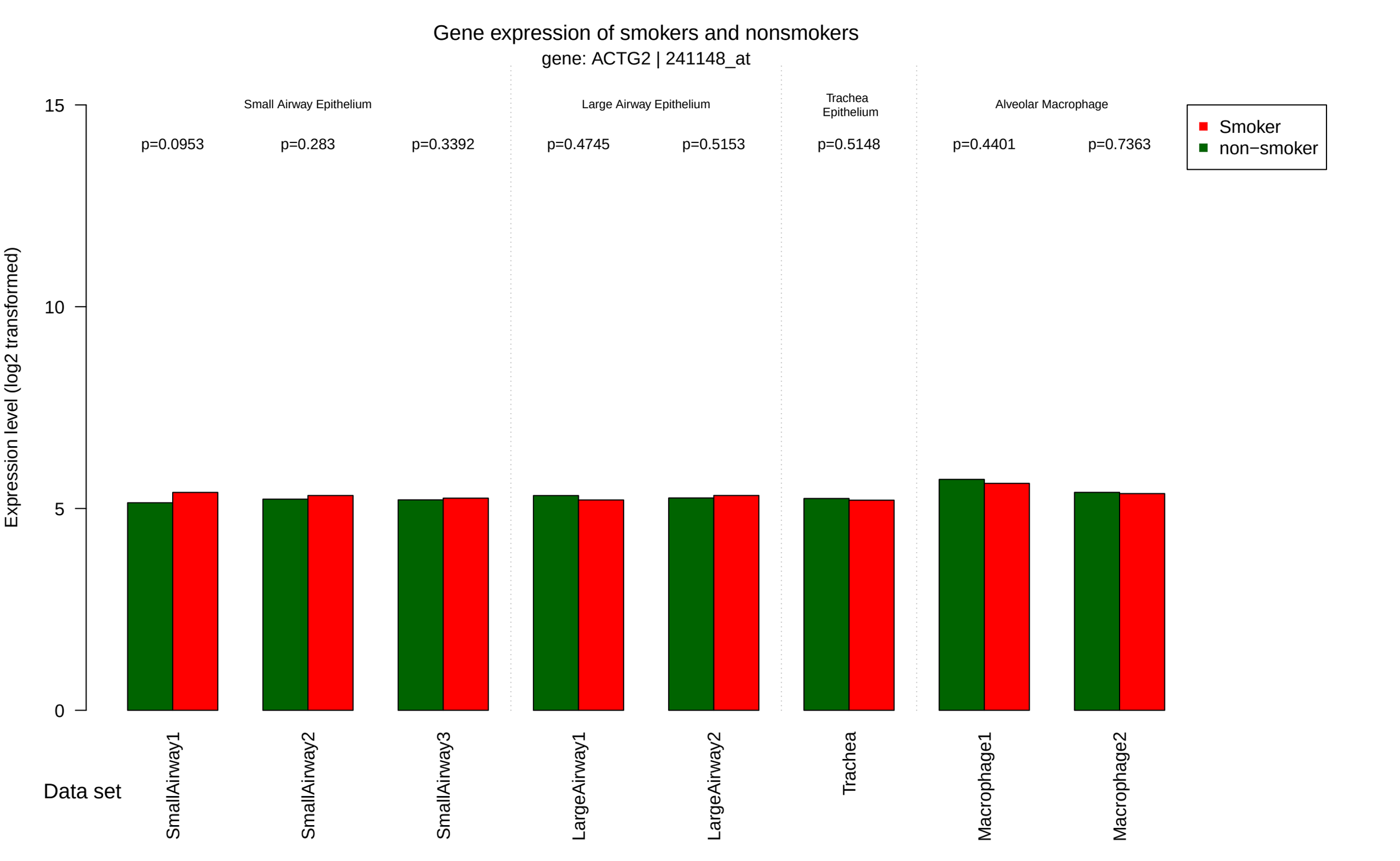
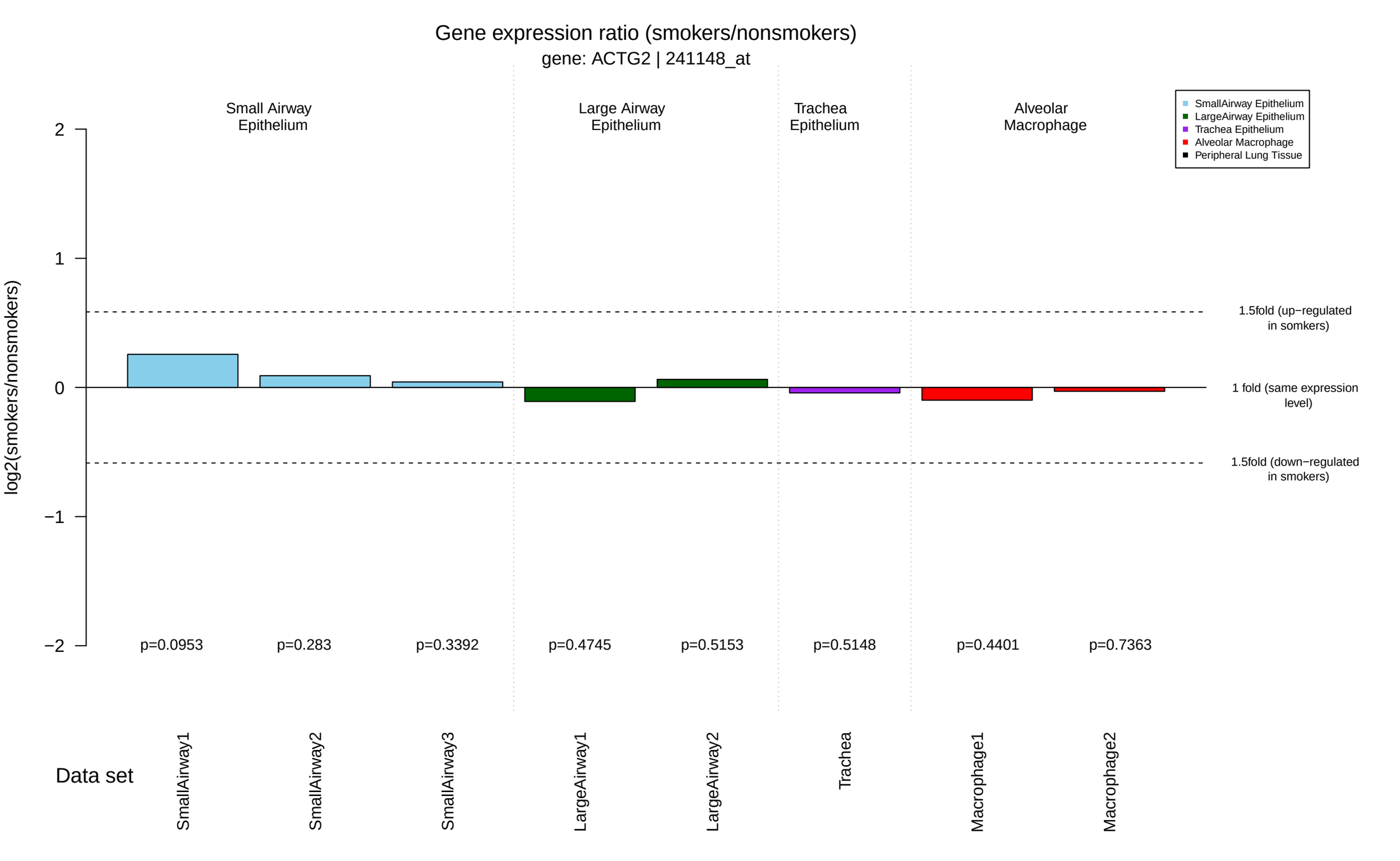
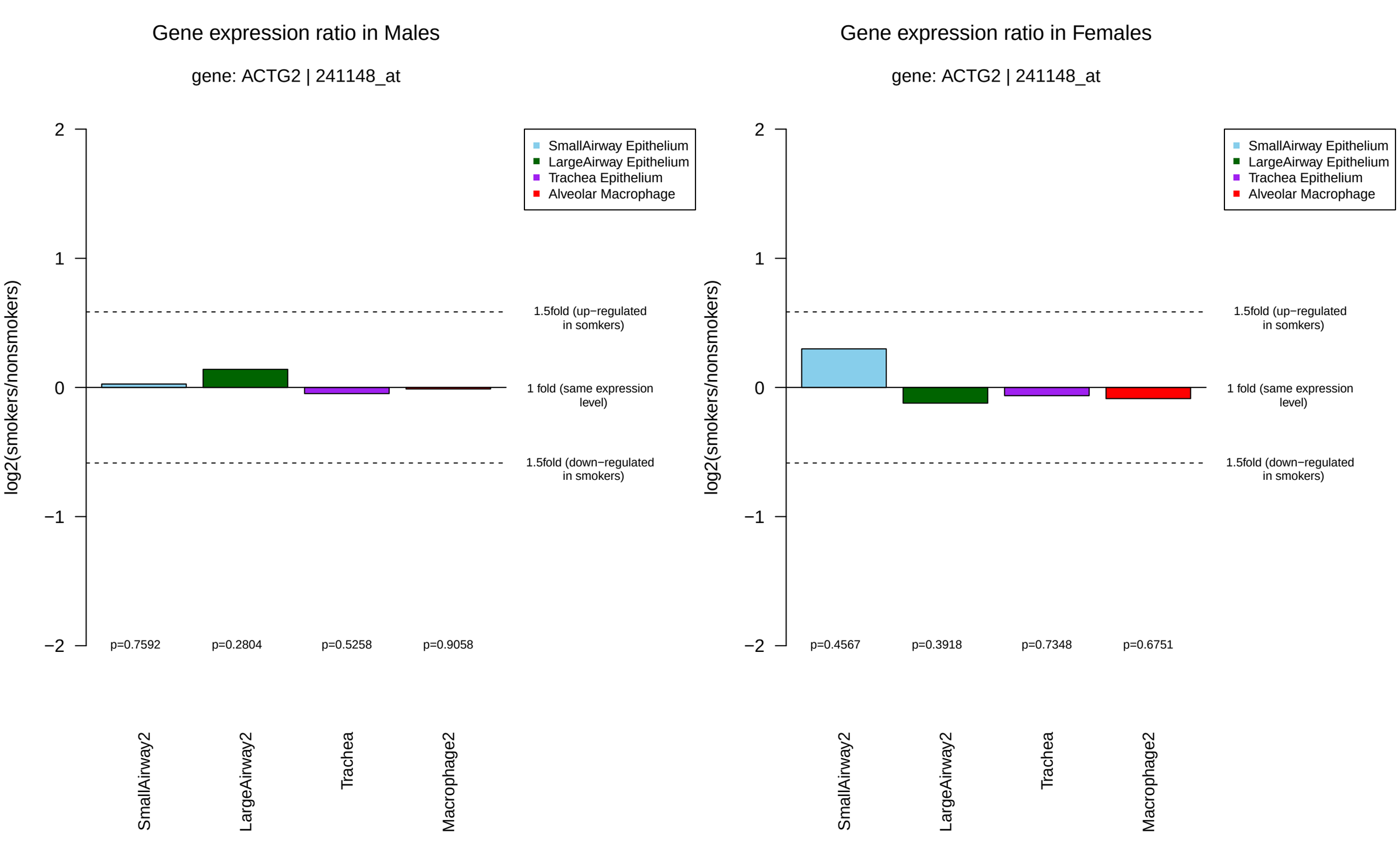
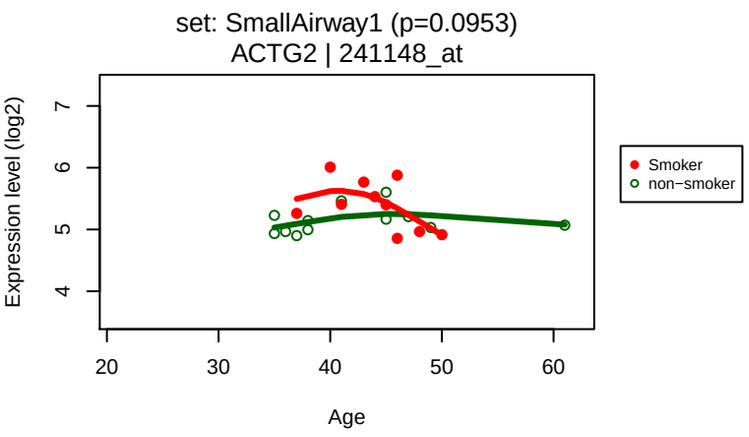
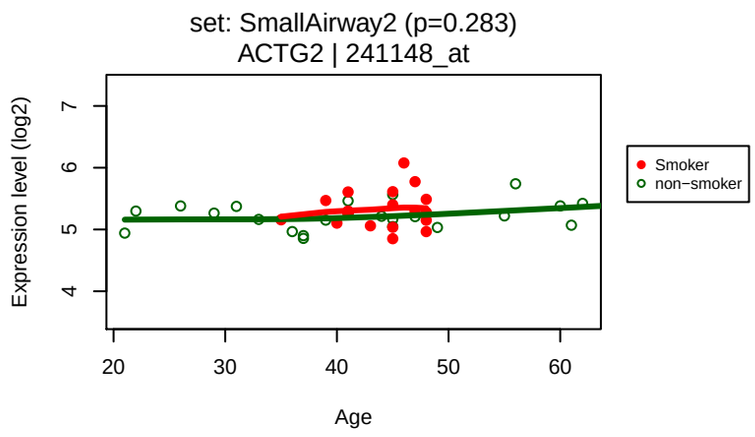
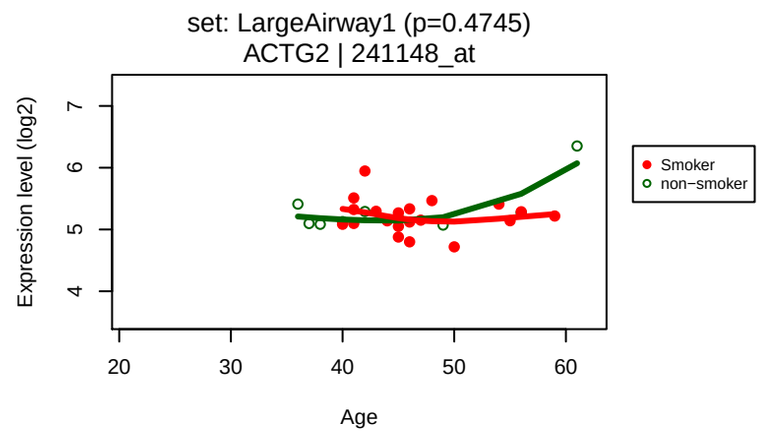

Gene expression of smokers and nonsmokers
gene: ACTG2 | 241148_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0953
p=0.283
p=0.3392
p=0.4745
p=0.5153
p=0.5148
p=0.4401
p=0.7363
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

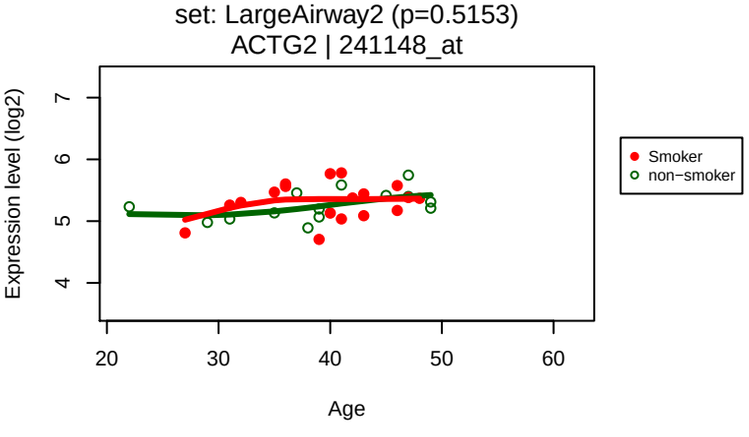

Gene expression ratio (smokers/nonsmokers)
gene: ACTG2 | 241148_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0953
p=0.283
p=0.3392
p=0.4745
p=0.5153
p=0.5148
p=0.4401
p=0.7363
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

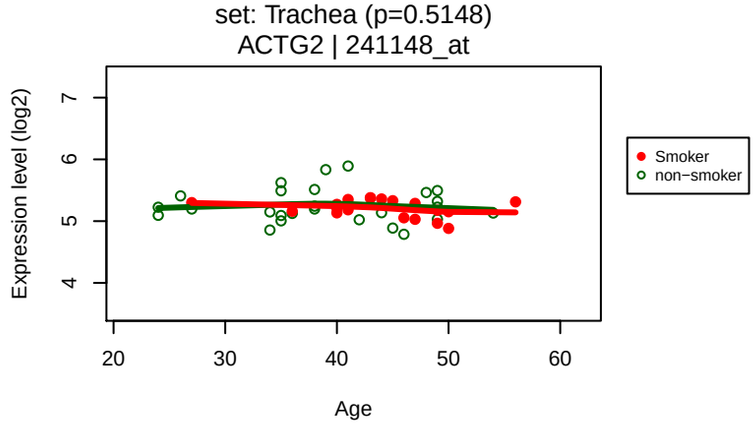

Gene expression ratio in Males
gene: ACTG2 | 241148_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7592
p=0.2804
p=0.5258
p=0.9058
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTG2 | 241148_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4567
p=0.3918
p=0.7348
p=0.6751
SmallAirway2
LargeAirway2
Trachea
Macrophage2

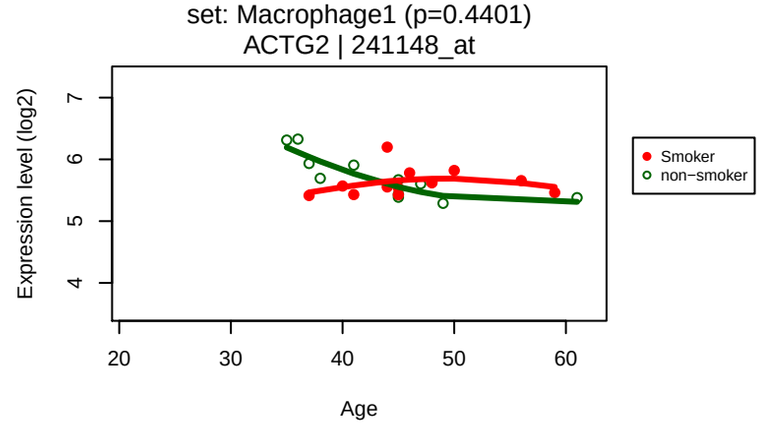

set: SmallAirway1 (p=0.0953)
ACTG2 | 241148_at
Expression level (log2)
Age
Smoker
non−smoker

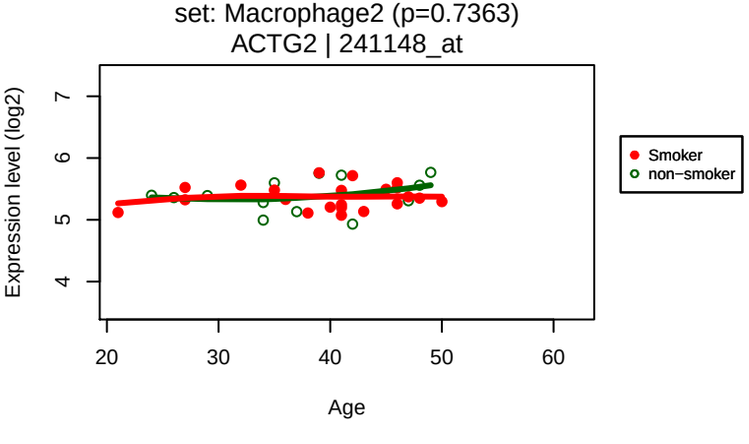

set: SmallAirway2 (p=0.283)
ACTG2 | 241148_at
Expression level (log2)
Age
Smoker
non−smoker

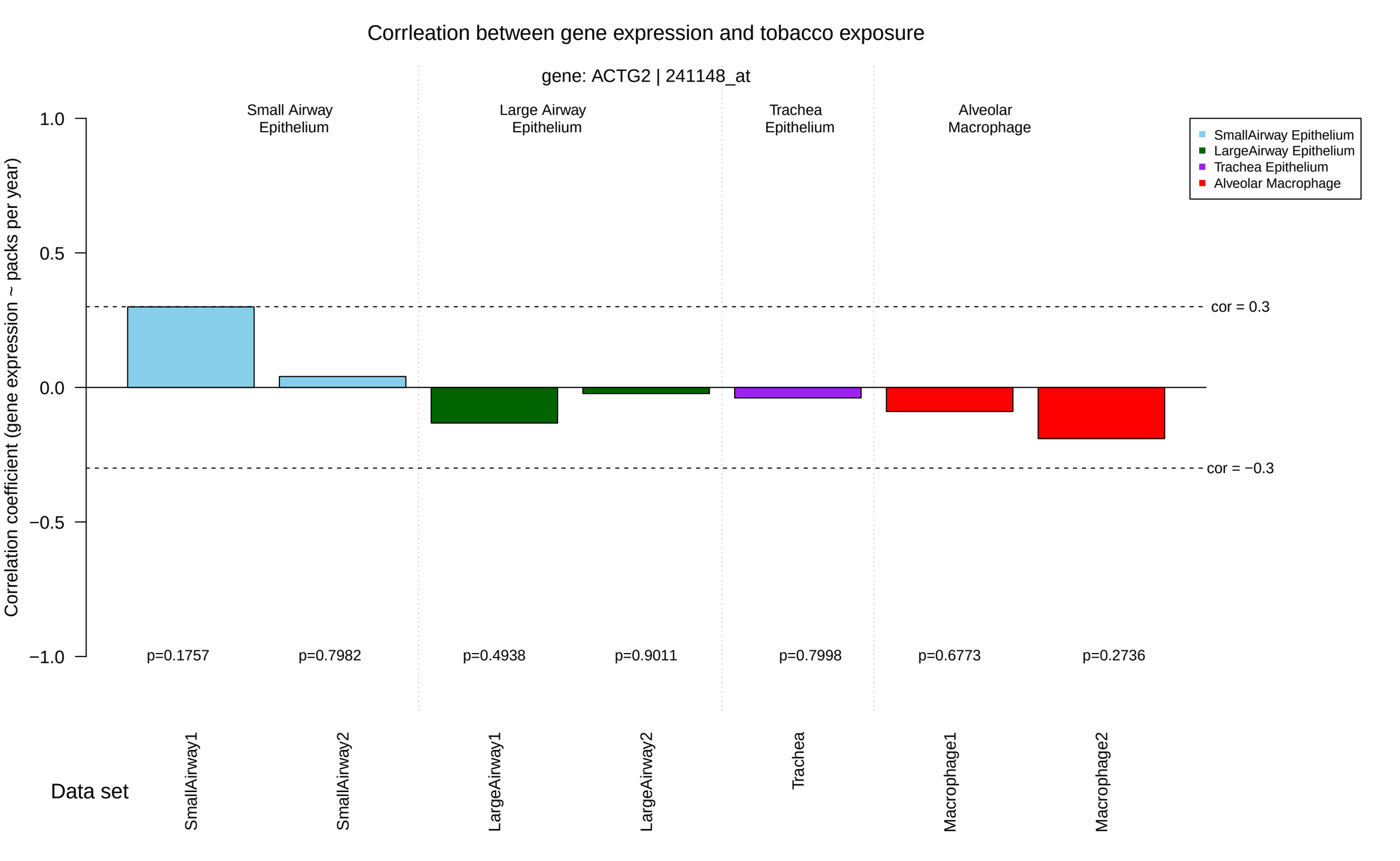

set: LargeAirway1 (p=0.4745)
ACTG2 | 241148_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5153)
ACTG2 | 241148_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5148)
ACTG2 | 241148_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4401)
ACTG2 | 241148_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7363)
ACTG2 | 241148_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTG2 | 241148_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1757
p=0.7982
p=0.4938
p=0.9011
p=0.7998
p=0.6773
p=0.2736
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2